



Emergence Of The Griffiths Phase Through The Application Of The Greenberg–Hastings Model To Connectome Networks

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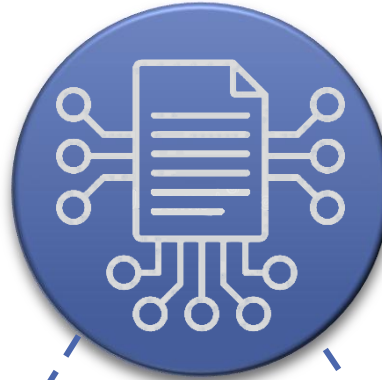


OUTLINE

Brain criticality hypothesis
Power-law data distribution



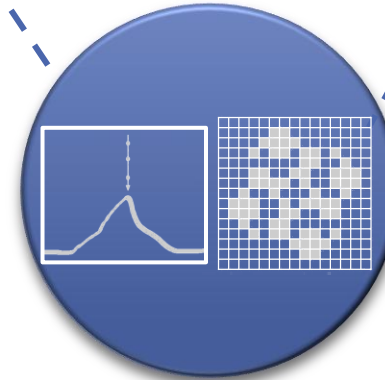
Connectome data



Results and Observations



How to reach critical state?
✓ Self-organized criticality
✓ Griffiths phase

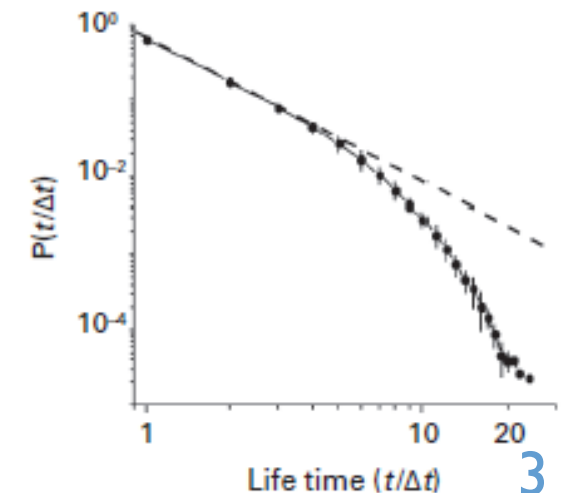
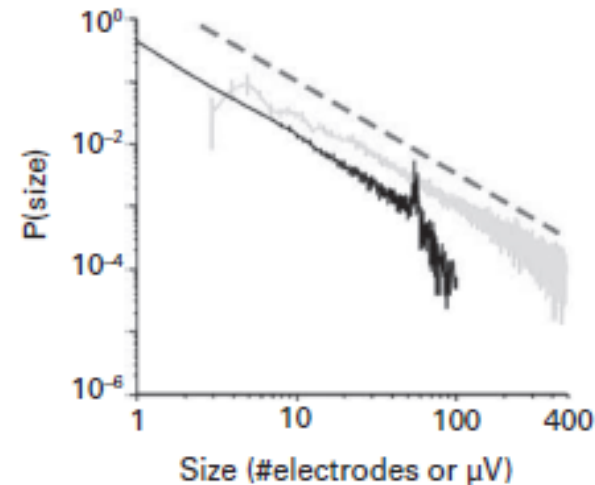
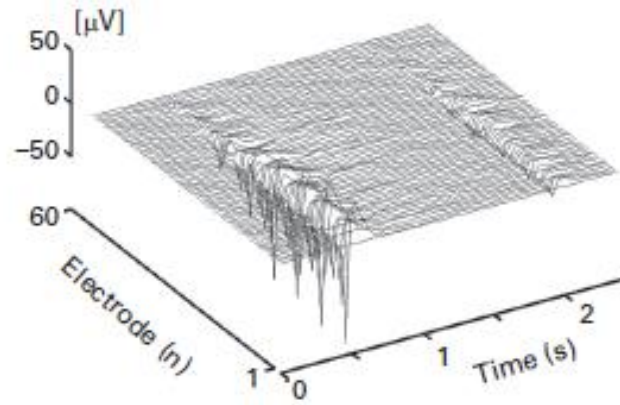
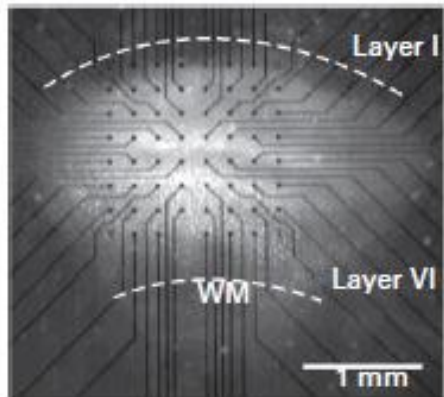


Sandpile model
Greenberg-Hastings model

■ BRAIN CRITICALITY HYPOTHESIS



- Observations such as power-law behavior of neural avalanches statistics
→ **Halmark of criticality**
- Example: Beggs and Plenz, 2003

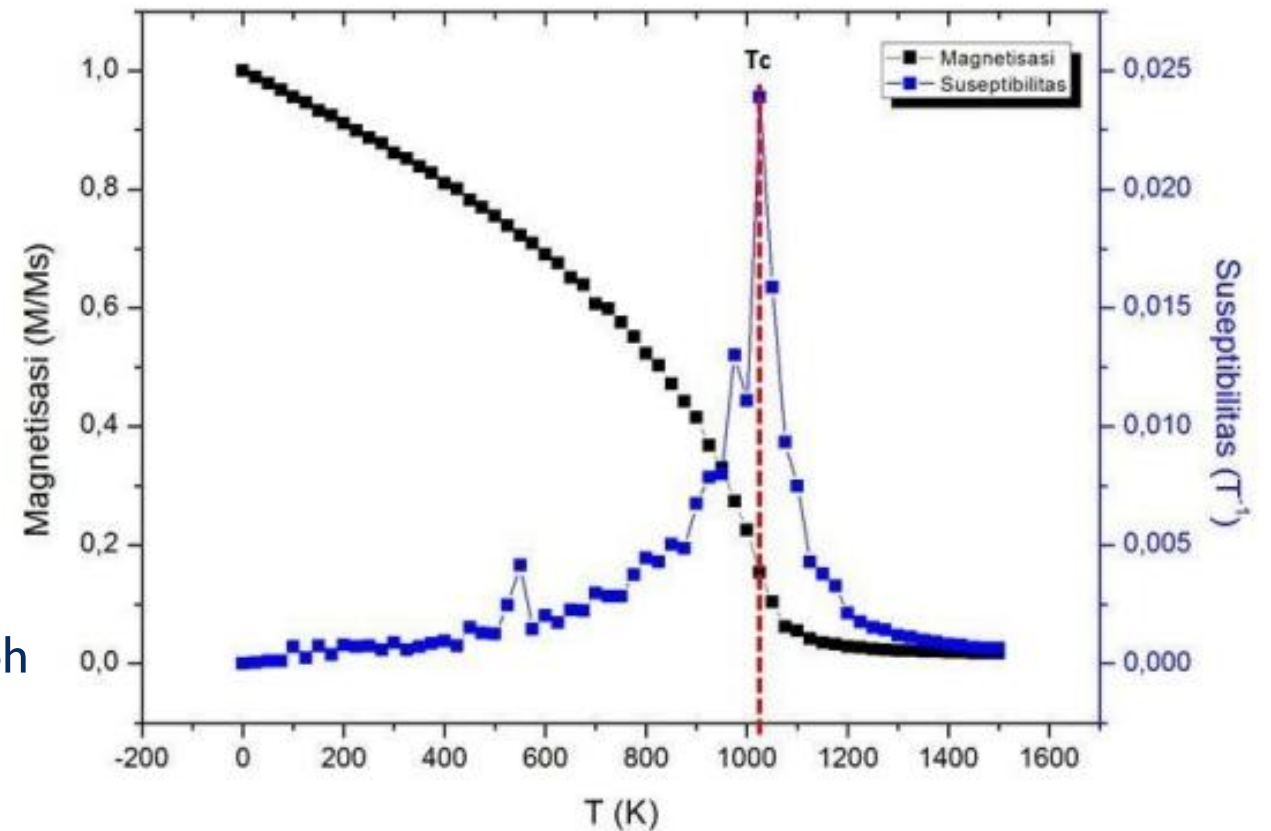


■ CRITICALITY IN PHYSICS

Curie Temperature:

The specific critical temperature for ferromagnetic materials, marking the transition from ferromagnetism to paramagnetism.

- curie temperature value could be determined by using magnetic susceptibility to temperature graph



Magnetization and susceptibility over temperature
(Rohman et al, 2021)

■ CRITICALITY IN PHYSICS

- T: 0 K, spin oriented parallel
- T: 200 K, the direction of rotation becomes a little random
- T: 600 K, more random
- T: Above curie temperature, spin orientation becomes totally random

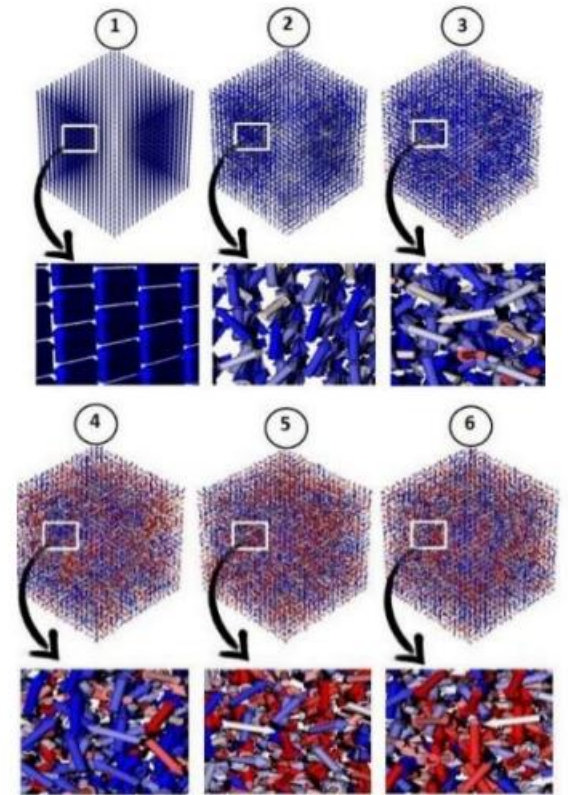
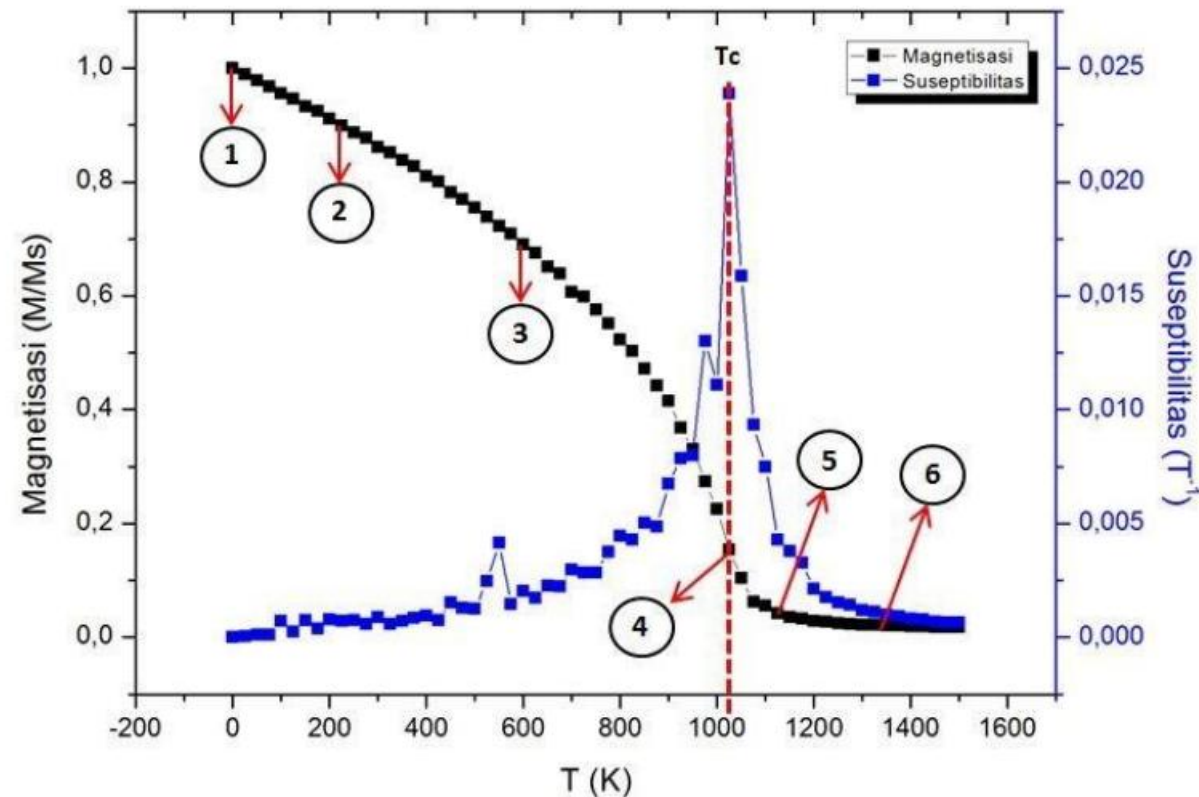


Figure 4. Spin orientation changing of Co₁₀Fe₉₀ alloy nanocube.

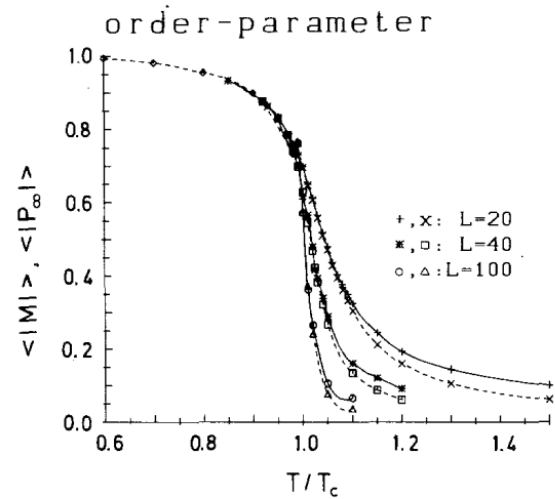


Fig. 3. Magnetization (—) and percolation probability (---) plotted vs. T/T_c for three system sizes: $\langle |M| \rangle$ for (+) $L=20$, (*) $L=40$, and (○) $L=100$, and P_∞ is for (×) $L=20$, (□) $L=40$, and (△) $L=100$.

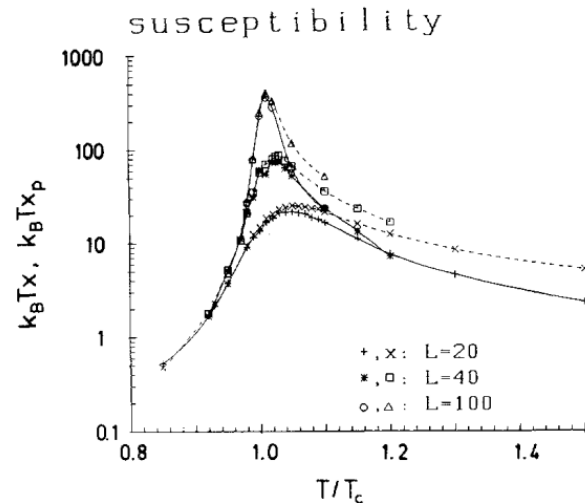
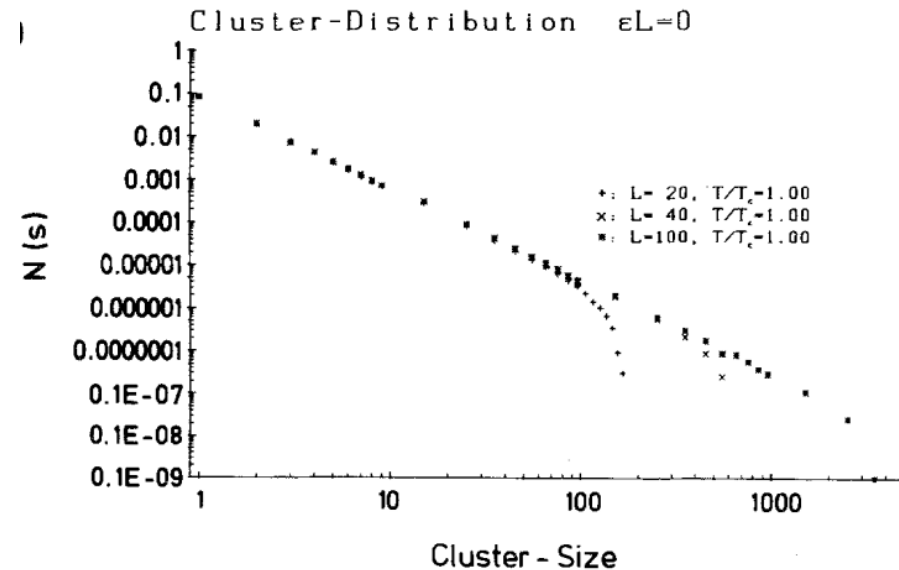


Fig. 5. Susceptibility $k_B T \chi'$, $k_B T \chi^{(p)}$ (on a logarithmic scale) plotted vs. T/T_c for three system sizes. Full curves refer to χ' and broken curves to $\chi^{(p)}$.

CRITICALITY IN PHYSICS

“Monte Carlo study of the Ising model phase transition in terms of the percolation transition of physical clusters”
D’Onorio De Meo, Heermann & Binder (1990)



■ HOW IS BRAIN TUNED TO REACH A CRITICAL POINT?



- Self-organized criticality : No external fine-tuning needed



- Griffiths phase : Critical behavior over a range of control parameter

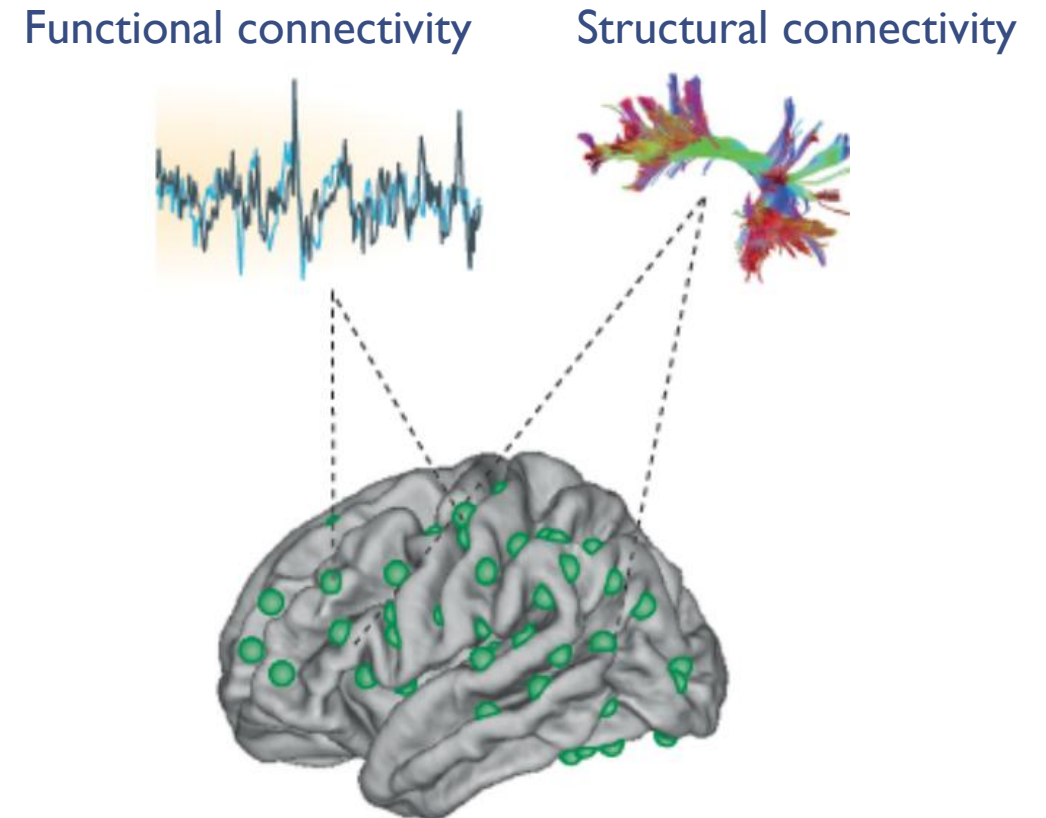


instead of an exact point, observed in heterogeneous networks

➤ Connectome data

■ CONNECTOME

- A comprehensive structural description of the network of elements and connections (Sporns et al, 2005)



- DATA: *Drosophila*

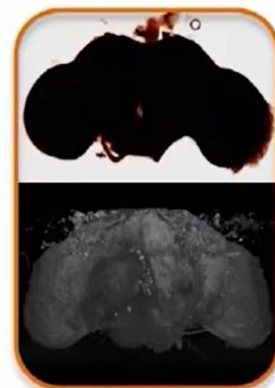


Cell

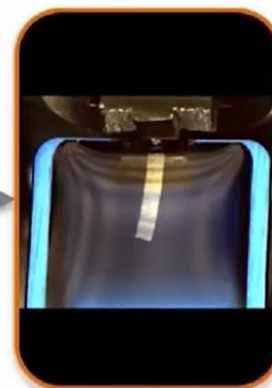
A Complete Electron Microscopy Volume of the Brain of Adult *Drosophila melanogaster*

2018

Zhihao Zheng, J. Scott Lauritzen, Eric Perlman, ..., Stephan Saalfeld, Richard D. Fetter, Davi D. Bock



Sample preparation and screening



Serial sectioning of whole brain

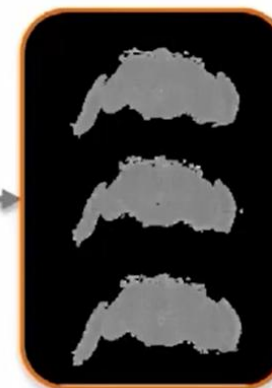
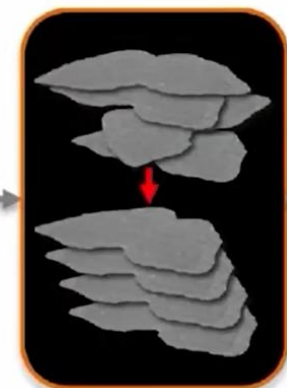


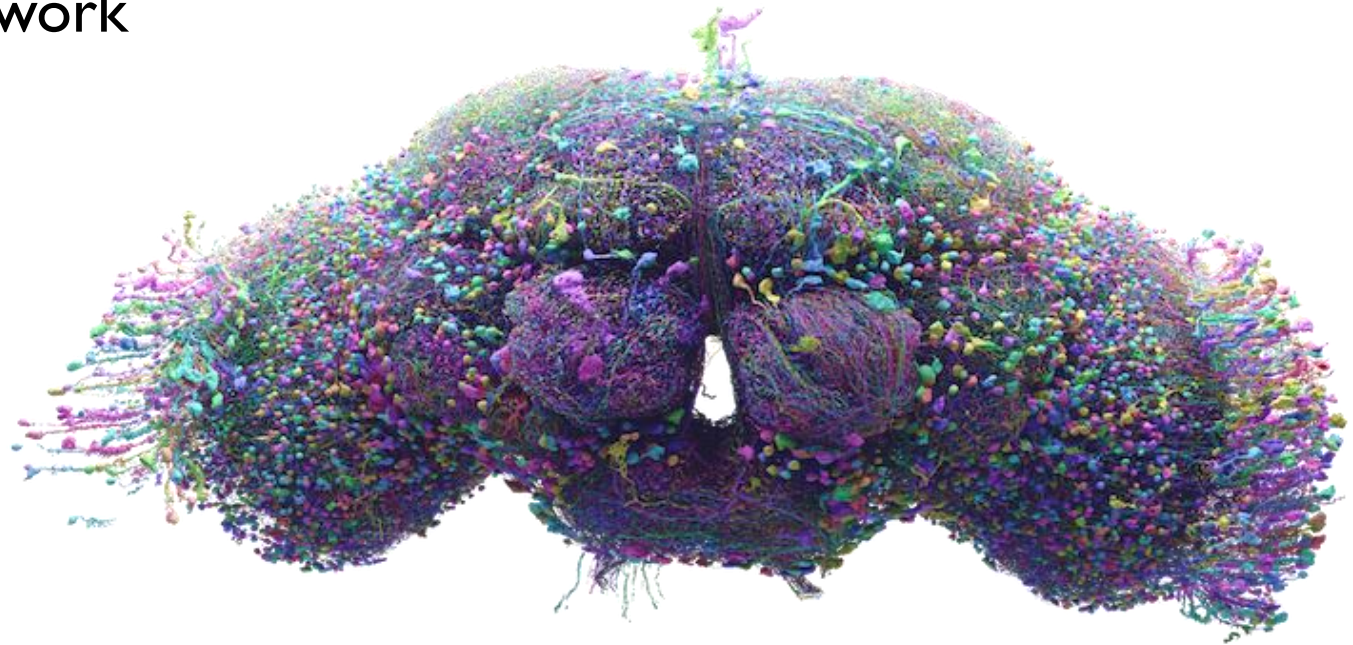
Image acquisition



Volume assembly

■ DATA: Drosophila

- A directed and weighted network
- 139,255 nodes
- 3,732,460 edges
- 50,666,648 synapses
- Each node : a neuron
- Each edge : synapse existence
- edge weights : total number of synapses



■ DATA: Drosophila

■ DATA IN CSV FORMAT

✓ CONNECTIONS

	pre_root_id	post_root_id	neuropil	syn_count	nt_type
0	720575940625363947	720575940623224444	ME_L	12	GABA
1	720575940630432382	720575940618518557	ME_L	67	ACH
2	720575940627314521	720575940626337738	ME_L	10	GABA
3	720575940620280405	720575940620204726	ME_L	15	ACH
4	720575940636942447	720575940613789411	LA_L	1	GLUT
...	...	...	...	...	...
5342441	720575940610739918	720575940617129885	PLP_L	5	ACH
5342442	720575940653332385	720575940623905719	CRE_R	2	ACH
5342443	720575940622485887	720575940631031803	ME_R	9	ACH
5342444	720575940620020534	720575940636489140	SMP_L	2	ACH
5342445	720575940632648612	720575940626847765	GNG	19	GABA

Neuropil Guide



sic parvis magna
flywire.ai

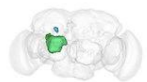


regions called **neuropils**.
Different neuropils are involved
in different processes:
Sight, flight, ...



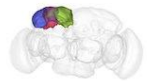
Optic Lobe

- ME (Medulla)
- LOP (Lobula Plate)
- LO (Lobula)
- AME (Accessory Medulla)
- LA (Lamina)
- OCG (Ocella)



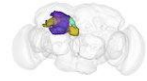
Lateral Complex

- LAL (Lateral Accessory Lobe)
- BU (Bulb)
- GA (Gall)



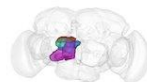
Superior Protocerebrum

- SMP (Sup. Medial Protocerebrum)
- SIP (Sup. Intermediate Protocerebrum)
- SLP (Sup. Lateral Protocerebrum)



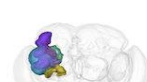
Inferior Protocerebrum

- ATL (Antler)
- CRE (Crepine)
- SCL (Superior Clamp)
- ICL (Inferior Clamp)
- IB (Inferior Bridge)



Ventromedial

- VES (Vest)
- SPS (Superior Posterior Slope)
- IPS (Inferior Posterior Slope)
- EPA (Epaulette)
- GOR (Gorget)



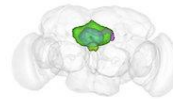
Ventrolateral

- AOTU (Anterior Optic Tubercle)
- AVLP (Anterior Ventrolateral Protocerebrum)
- PVLP (Posterior Ventrolateral Protocerebrum)
- PLP (Posterior Lateral Protocerebrum)
- WED (Wedge)



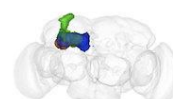
Central Complex

- EB (Ellipsoid Body)
- FB (Fan Body)
- PCB (Protocerebral Bridge)
- NO (Noduli)



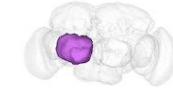
Mushroom Body

- MB_VL (Vertical Lobe)
- MB_ML (Medial Lobe)
- MB_PED (Pediculus)
- MB_CA (Calyx)



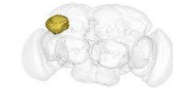
Antenna Lobe

- LAL (Lateral Accessory Lobe)



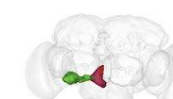
Lateral Horn

- LH



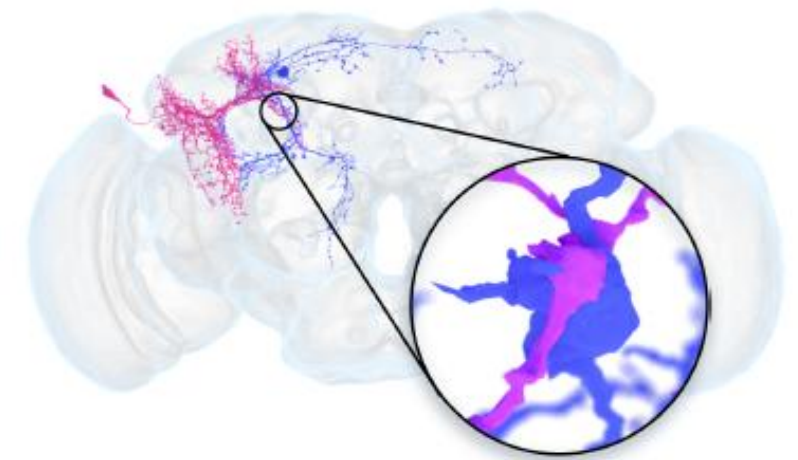
Periesophageal

- FLA (Flange)
- CAN (Cante)
- PRW (Prow)
- SAD (Saddle)
- AMMC (Antennal Mechanosensory and Motor Center)



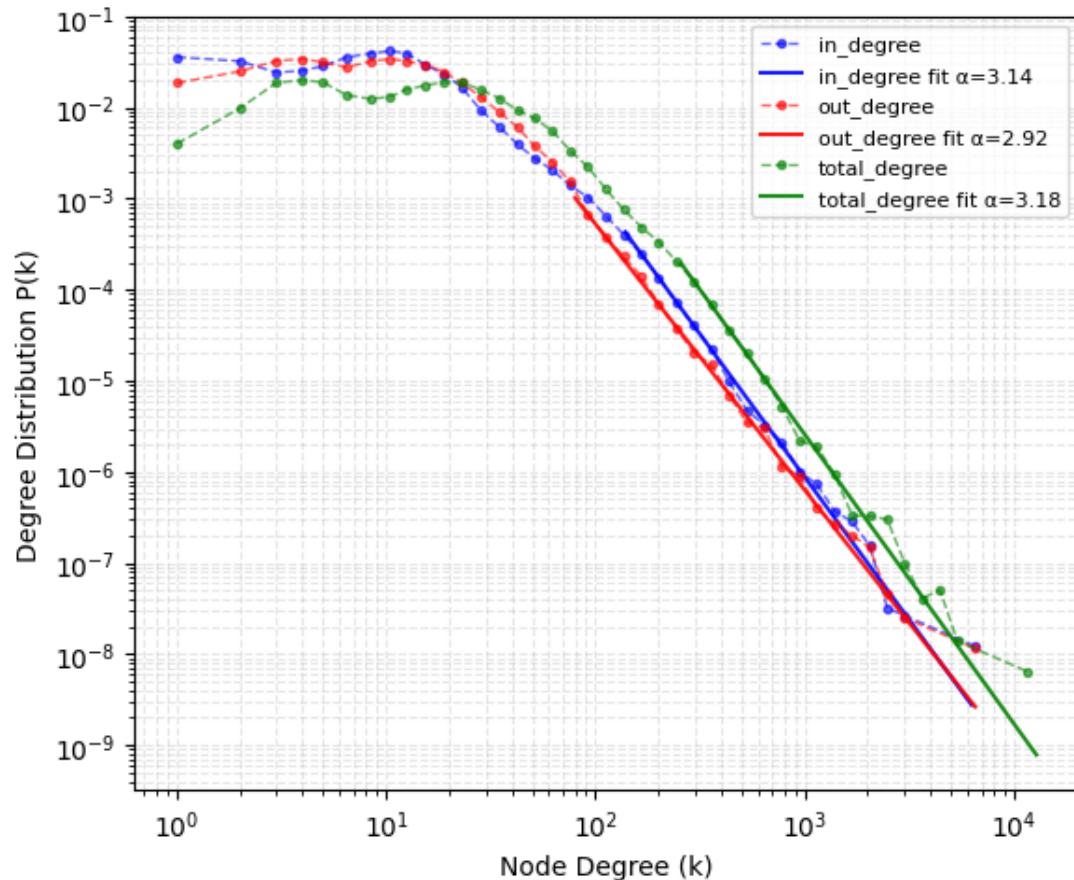
Gnathal Ganglia

- GNG (Gnathal Ganglia)



Connections and synapses

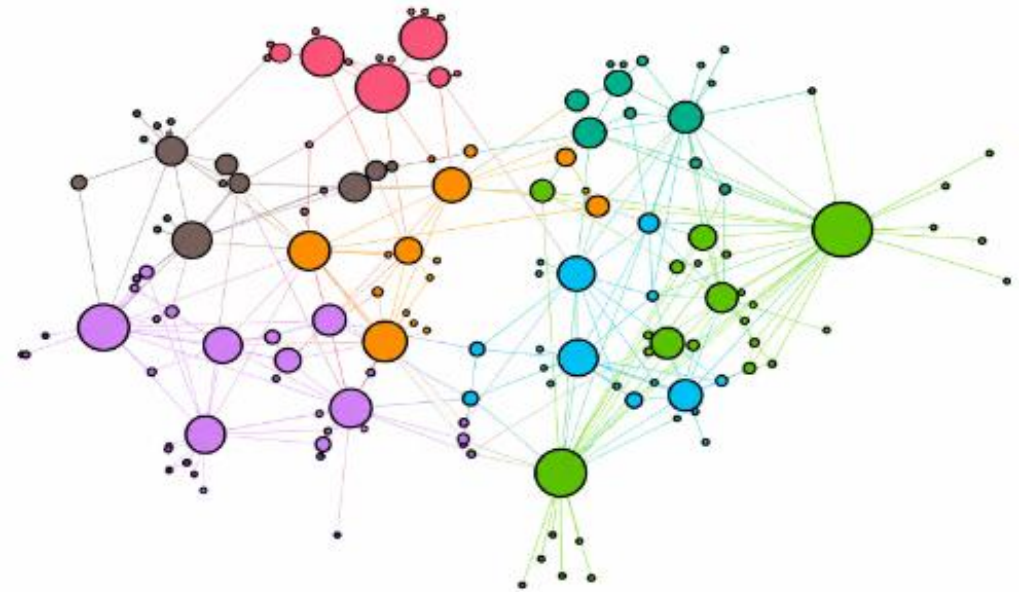
■ DATA: Drosophila



Stats	In-degree	Out-degree	Total degree
Mean	26.93	26.93	53.7
Std	60.74	54.51	108.41
Min	0	0	1
First quartile	7	9	17
Median	14	17	32
Third quartile	26	31	59
Max	6261	6523	12784

■ DATA: HUMAN CONNECTOME, KKI-18

- An **undirected** and **weighted** network
- 848,849 nodes
- 66,298,600 edges.
- Each **node** : 1 mm³ brain voxels
- Edge **weights** : number of white matter fiber tracts



*Modular structure of the
human connectome
(Odor and Kelling, 2019)*

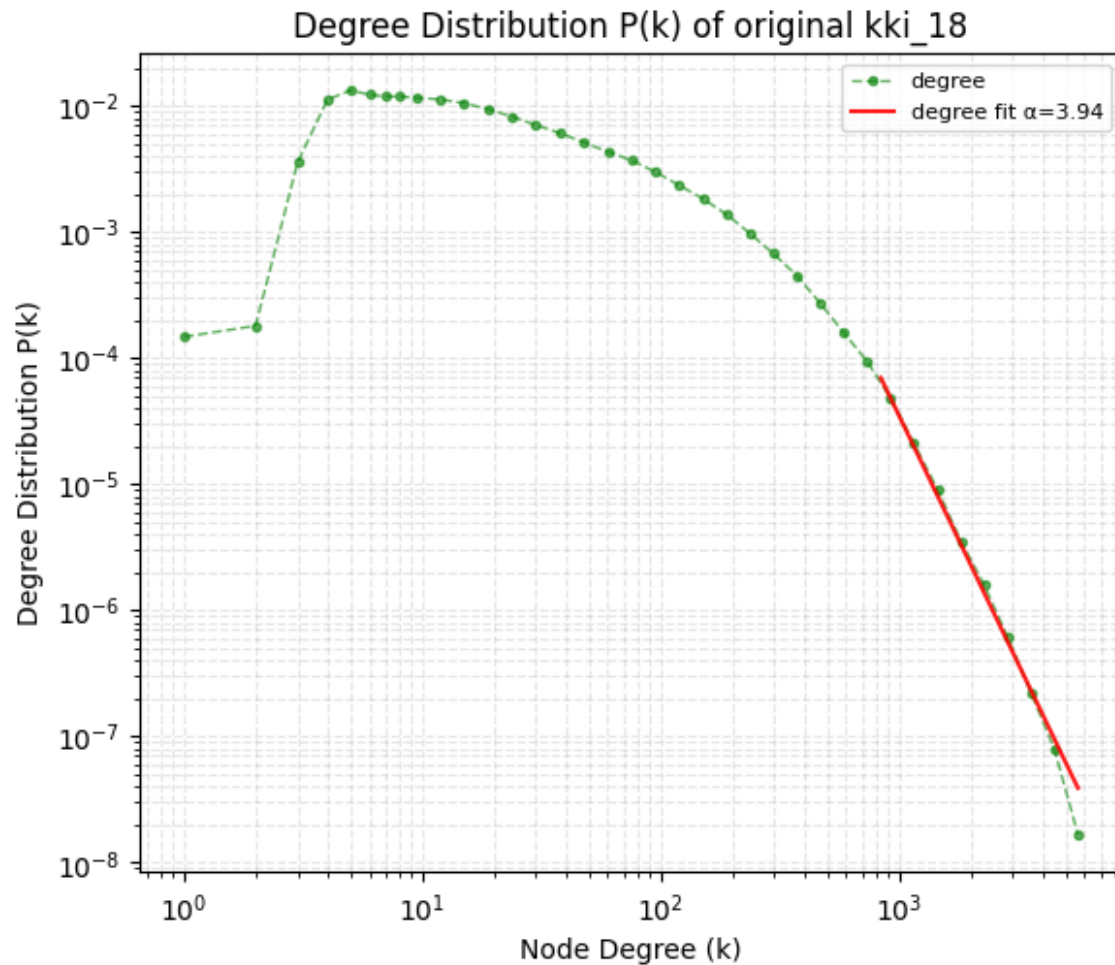
■ DATA: HUMAN CONNECTOME, KKI-18

■ DATA IN GRAPHML FORMAT

EDGES

```
<edge source="n135879" target="n196301">  
  <data key="e_weight">7</data>  
</edge>  
<edge source="n438900" target="n720958">  
  <data key="e_weight">3</data>  
</edge>  
<edge source="n85145" target="n92457">  
  <data key="e_weight">16</data>  
</edge>  
<edge source="n277147" target="n278907">  
  <data key="e_weight">1</data>  
</edge>  
<edge source="n311083" target="n314555">  
  <data key="e_weight">3</data>
```

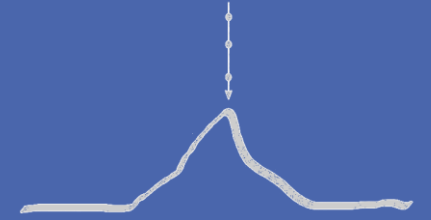
■ DATA: HUMAN CONNECTOME, KKI-18



Stats	Degree
Mean	156.27
Std	232
Min	1
First quartile	28
Median	77
Third quartile	190
Max	6291

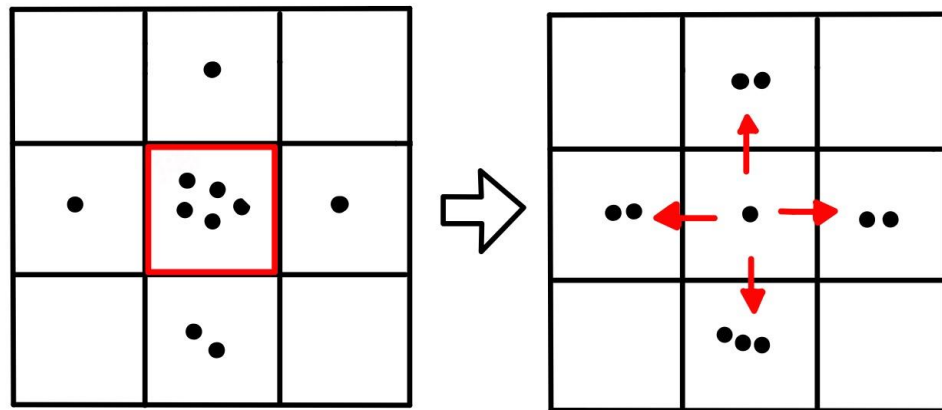
➤ MODELS & ➤ RESULTS

MODELS: Sandpile Model



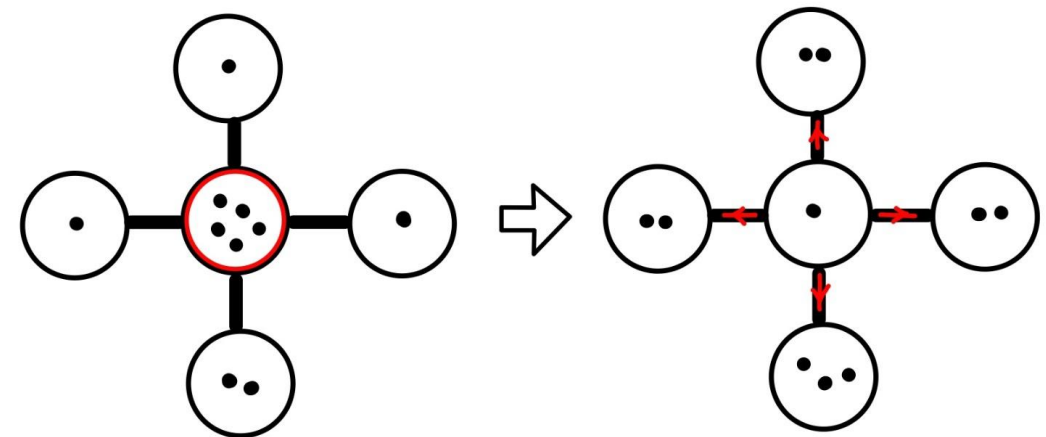
Classic sandpile

on a 2D grid

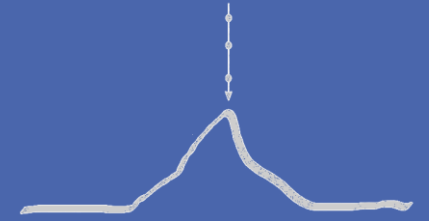


Or equivalently →

on a graph



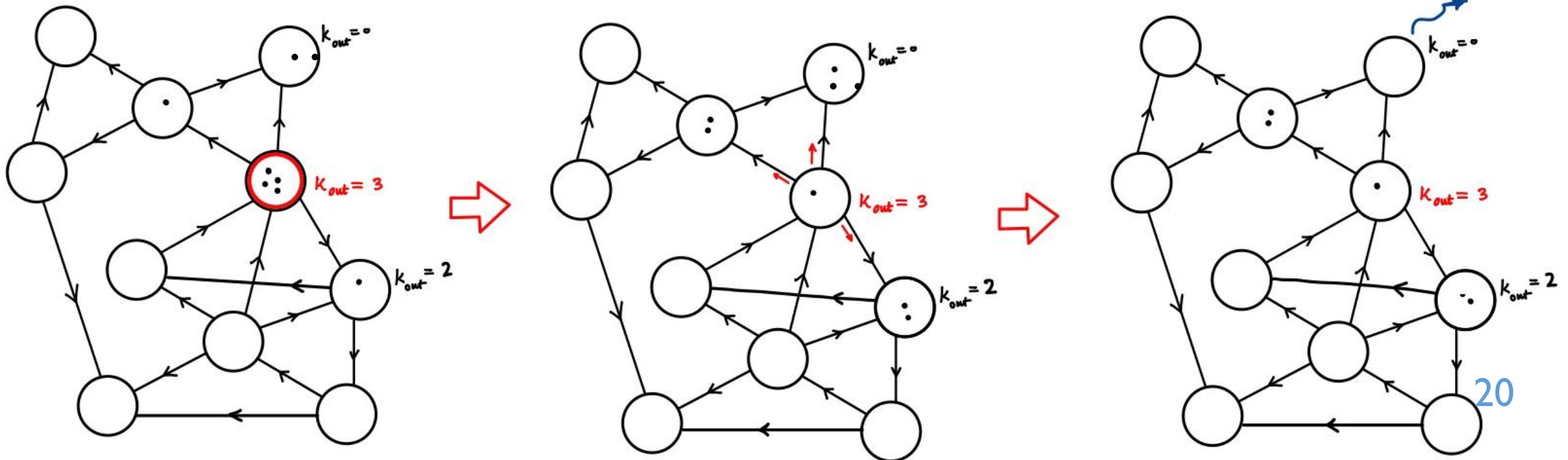
MODELS: Sandpile Model



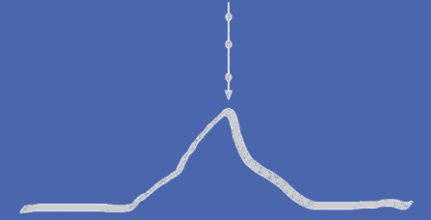
So, in our network:

Threshold?

Periphery nodes?



MODELS: Sandpile Model



$$k_i^{\text{eff}} = \max(k_{\text{out},i}, 3)$$

Threshold $\rightarrow$

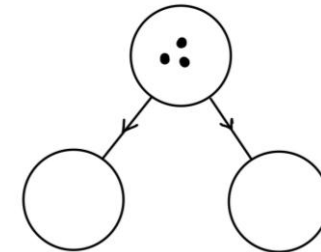
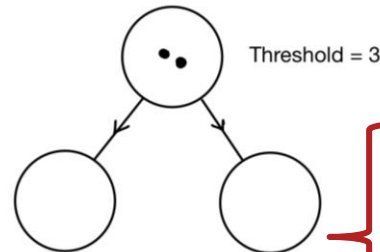
$$z_i^{\text{th}} = k_i^{\text{eff}}$$

Toppling
And
Dissipation

$$\rightarrow z_i \geq z_i^{\text{th}}$$

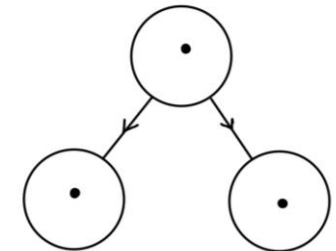
$$k_i = 0$$

$$k_i \in \{1, 2\}$$

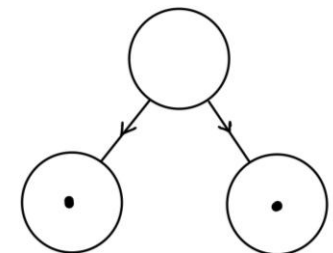


$$90\% : z_i \leftarrow k_i$$

$$10\% : z_i \leftarrow z_i^{\text{th}}$$



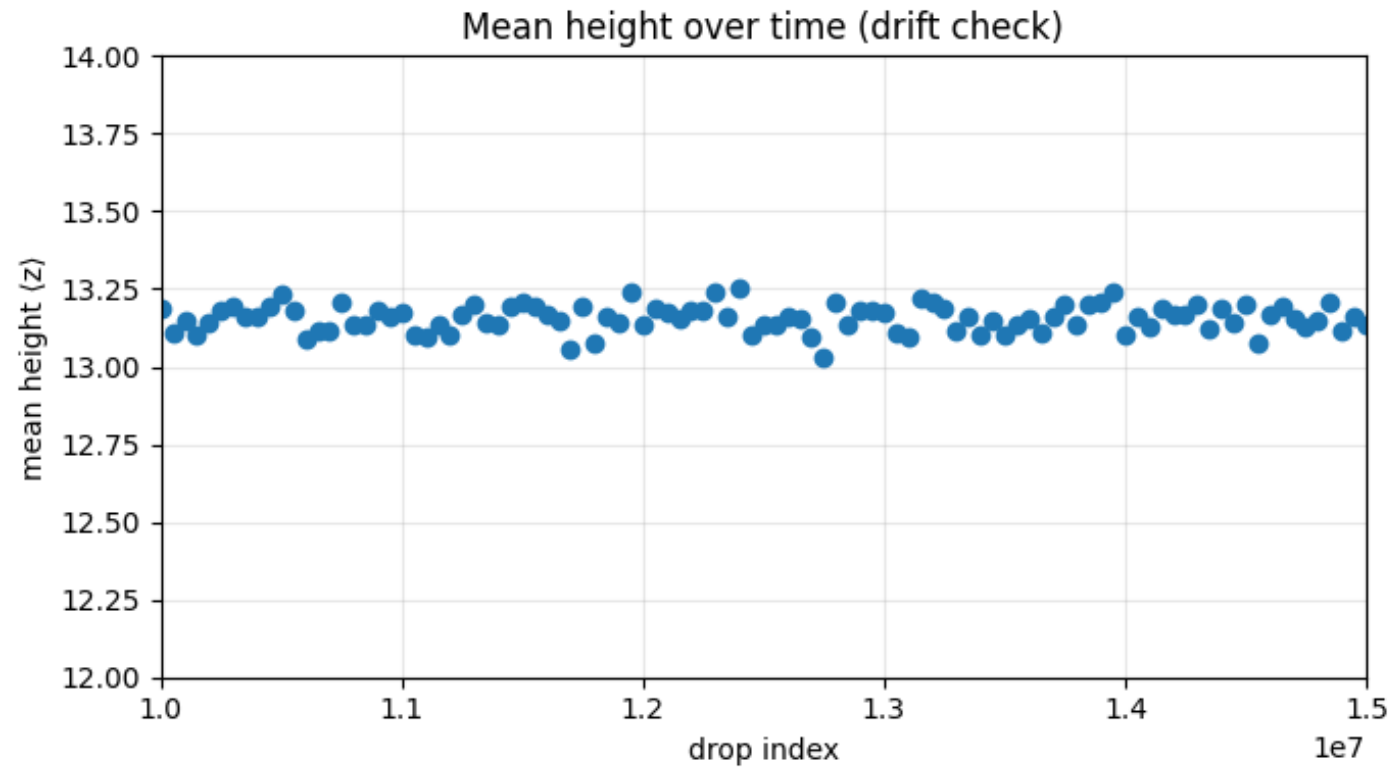
90 % : No dissipation



10 % : 1 grain is lost

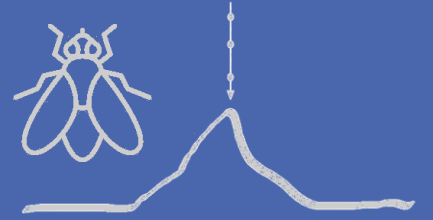
■ RESULTS: Sandpile Model

- Drosophila Connectome



■ RESULTS: Sandpile Model

➤ Drosophila Connectome



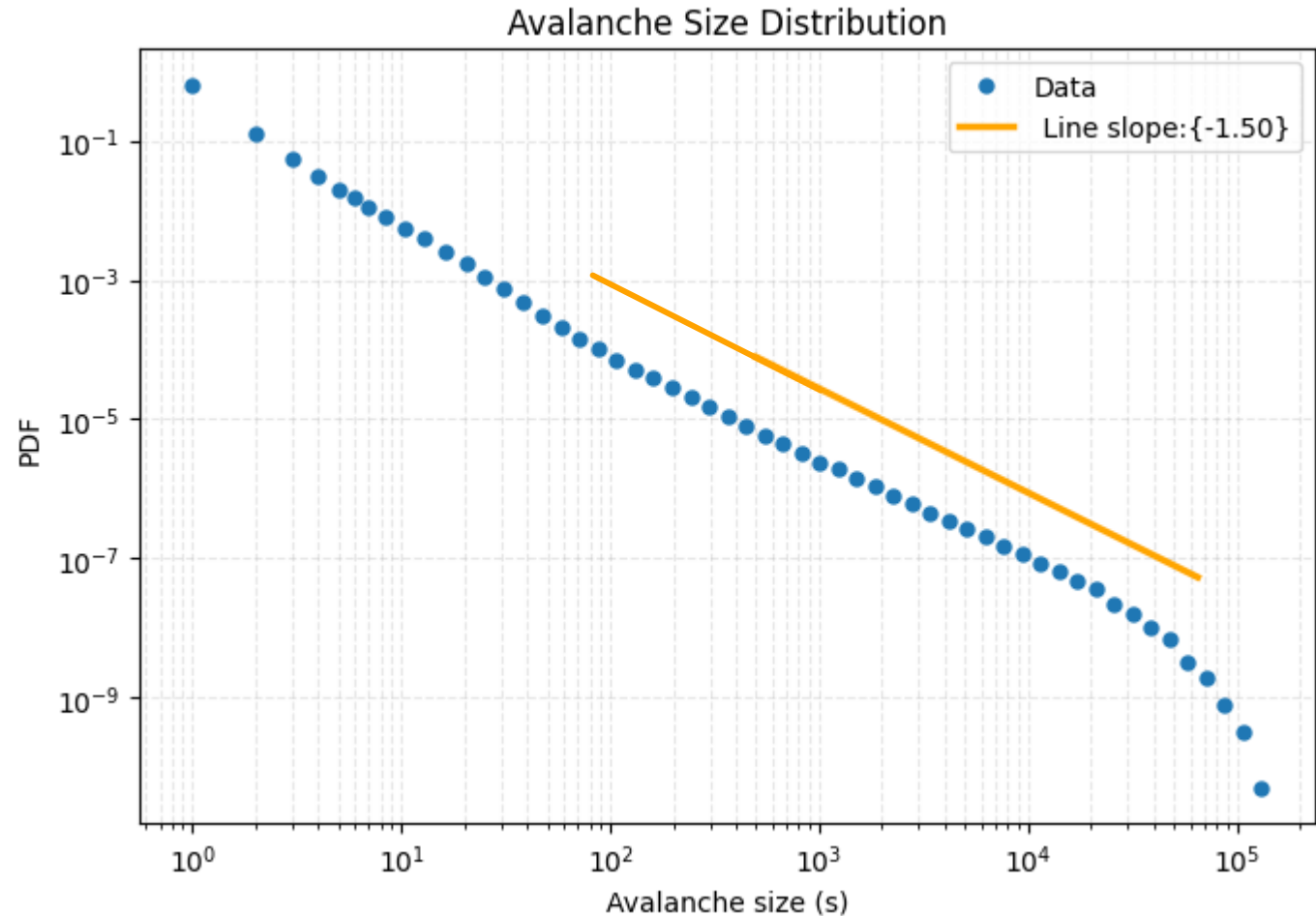
s : Avalanche size

$$\log P(s) \propto -\tau \log s$$

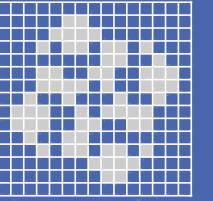
$$P(s) \sim s^{-\tau}, \tau \approx 1.46 \pm 0.03$$

Network	Slope	
2D Grid	1.2	
Erdos-Renyi	1.5	
Scale-free	$2 > \gamma > 3$	$\frac{\gamma}{\gamma - 1}$
	$\gamma > 3$	1.5

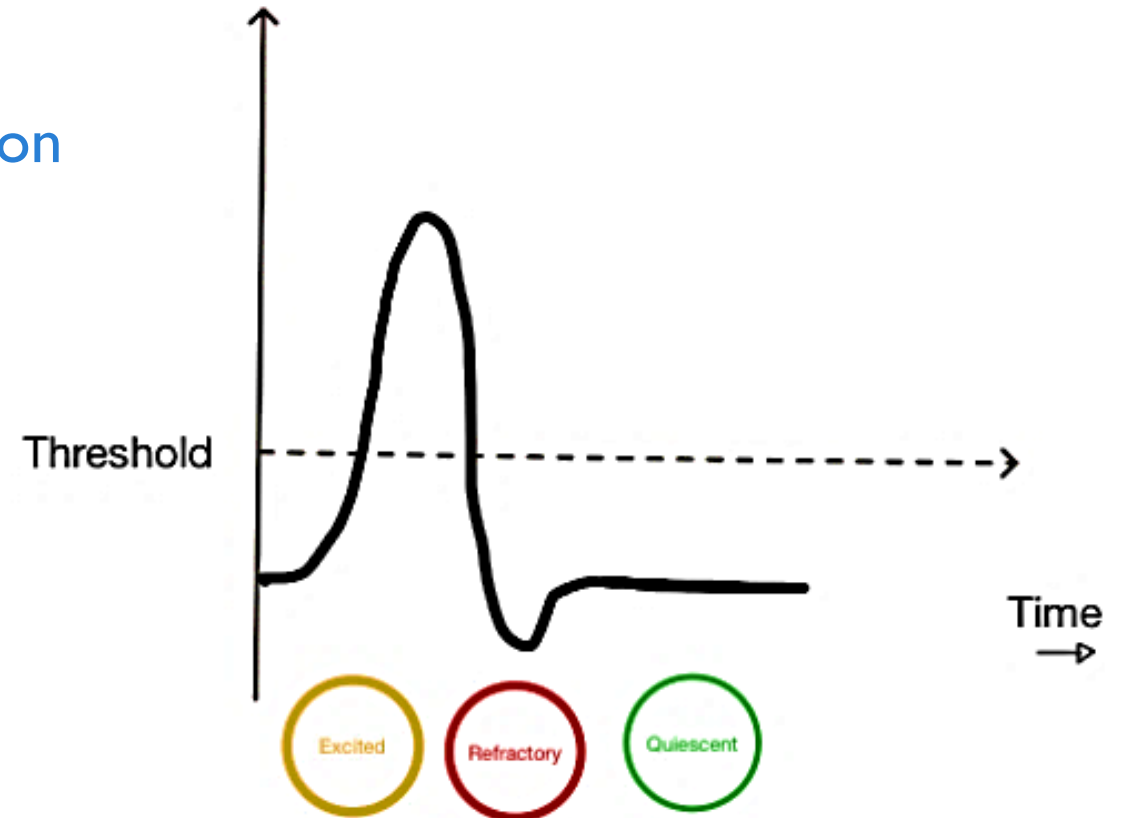
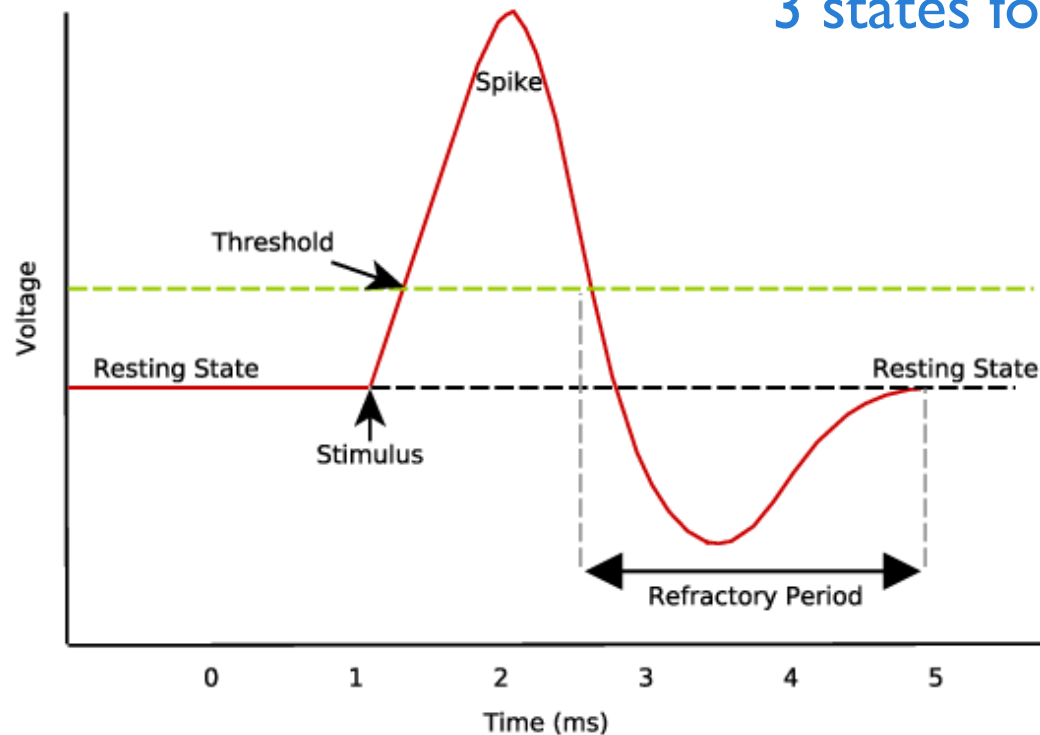
* γ : degree exponent



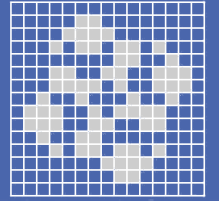
MODELS: Greenberg-Hastings Model



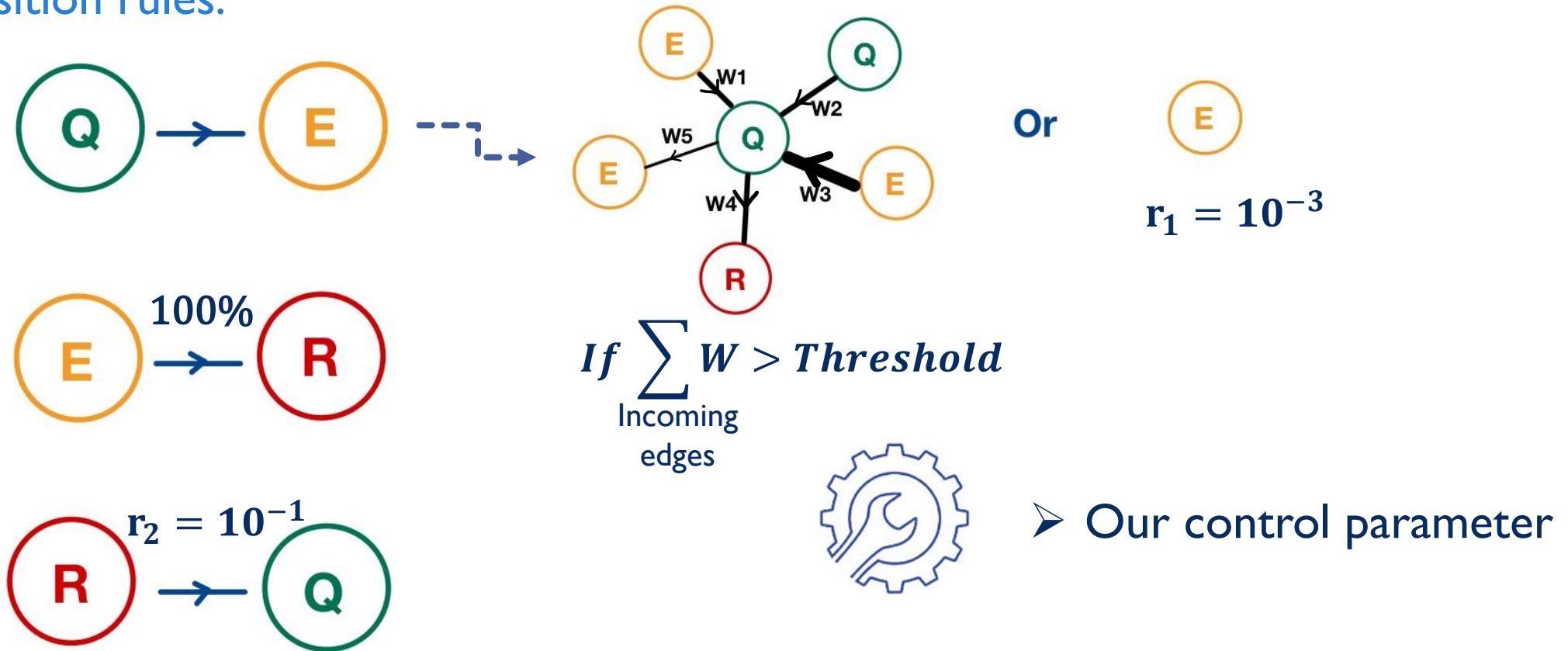
Modeling :
3 states for neuron



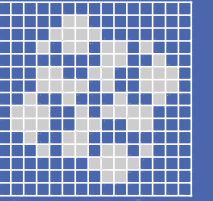
MODELS: Greenberg-Hastings Model



➤ Transition rules:



MODELS: Greenberg-Hastings Model



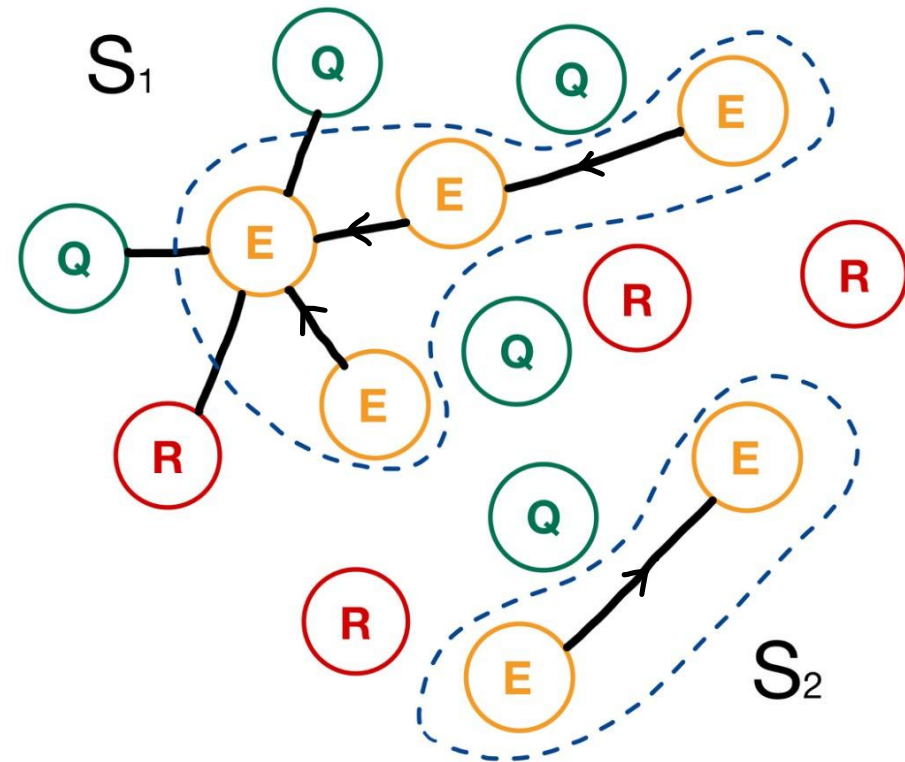
What is measured?

Cluster size:
Number of structurally
connected excited neurons

S_1 : largest cluster

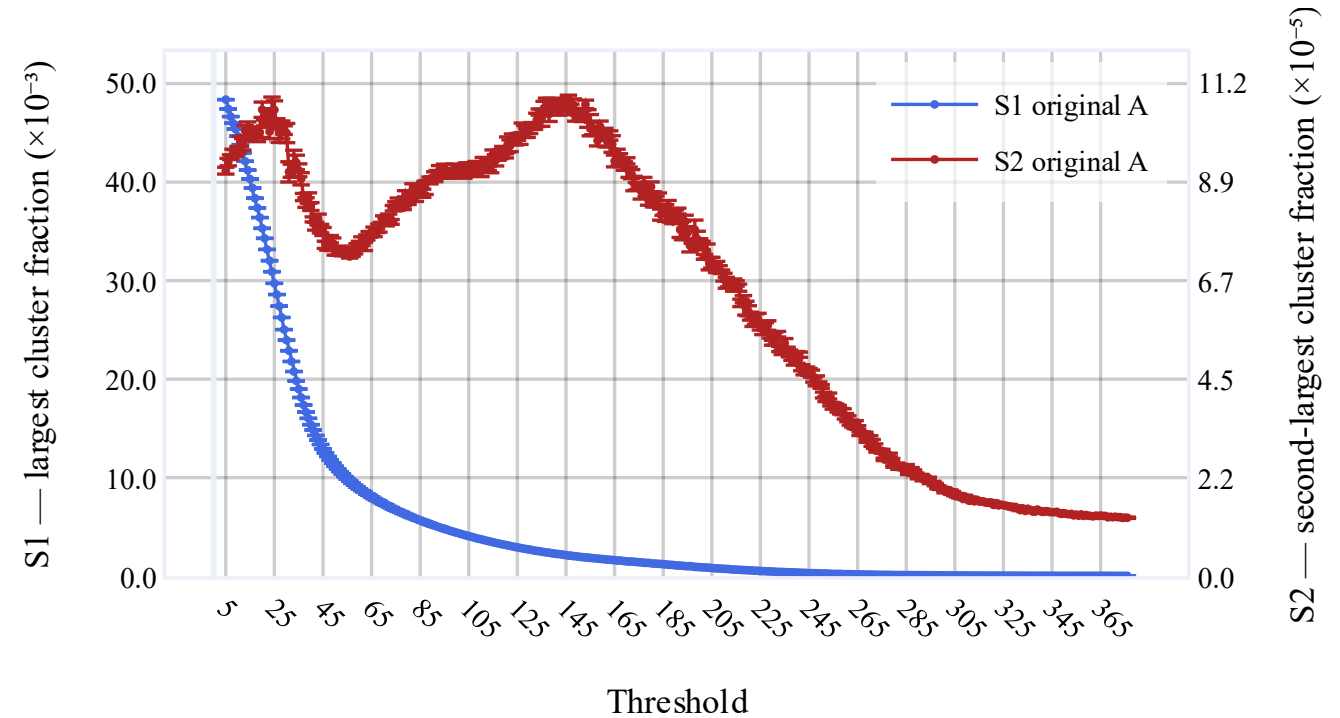
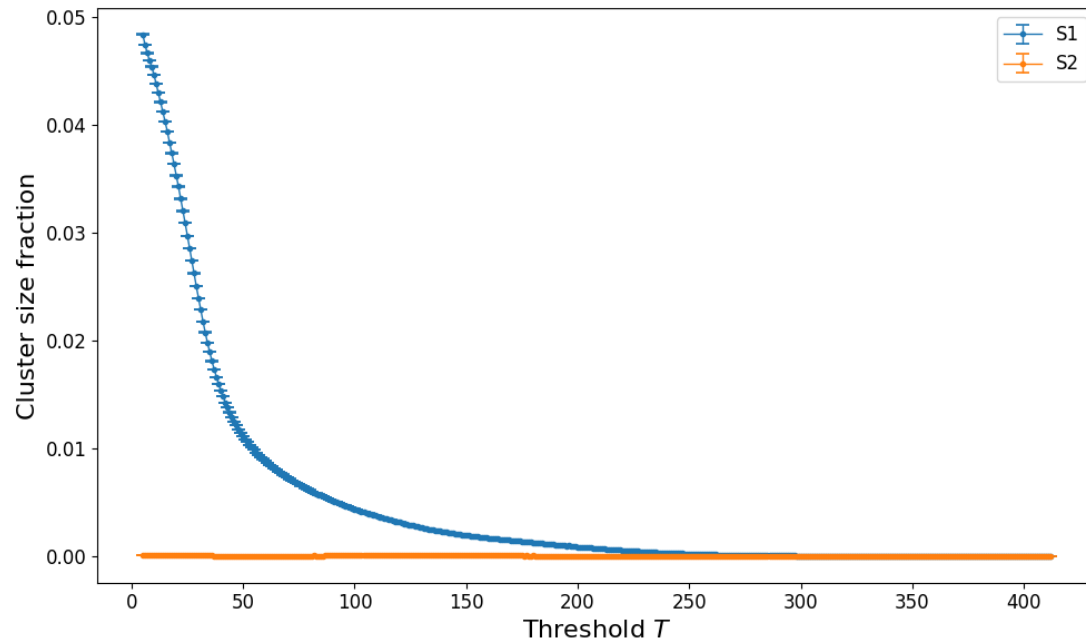
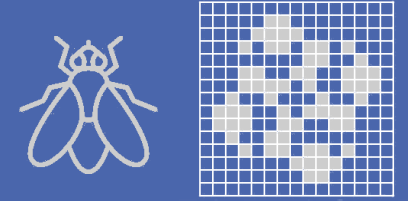
S_2 : second largest cluster

→ In each threshold

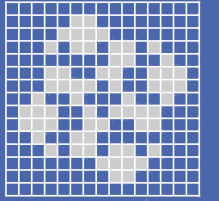


RESULTS: Greenberg-Hastings Model

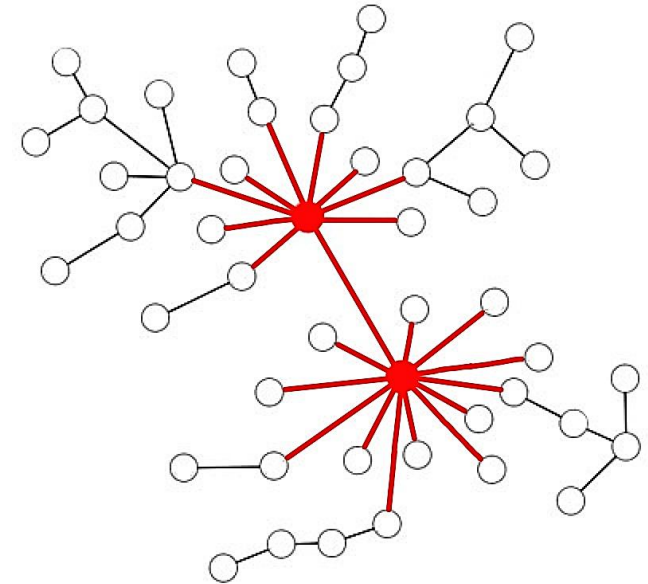
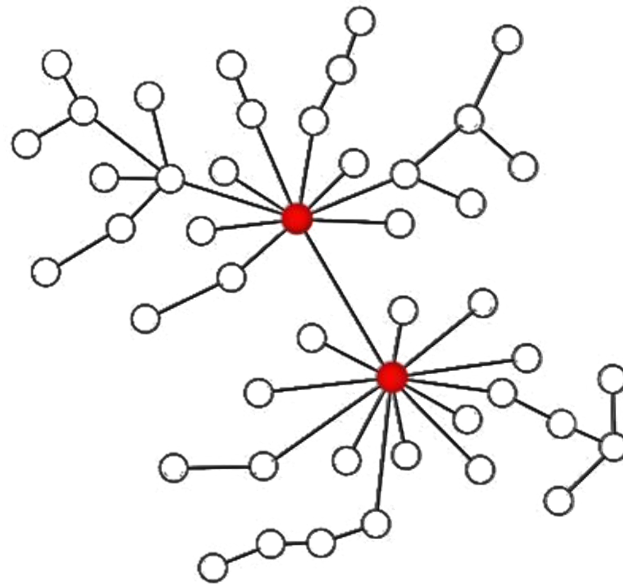
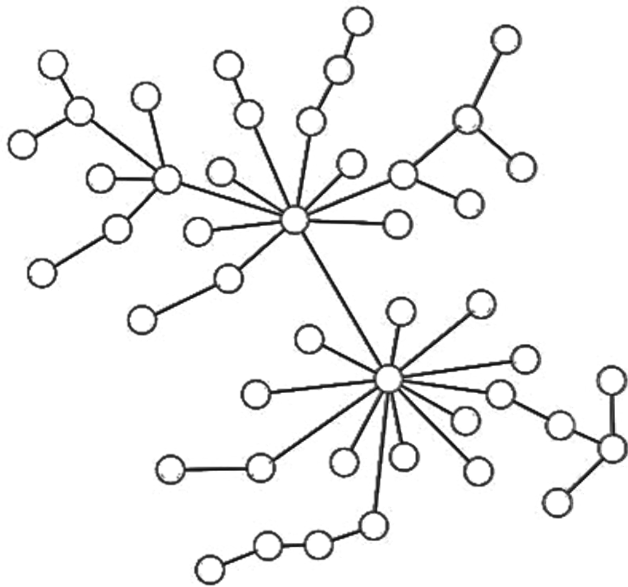
➤ Drosophila Connectome



■ RESULTS: Greenberg-Hastings Model

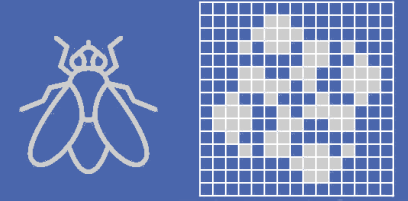


➤ Removing Hubs to Decrease Heterogeneity

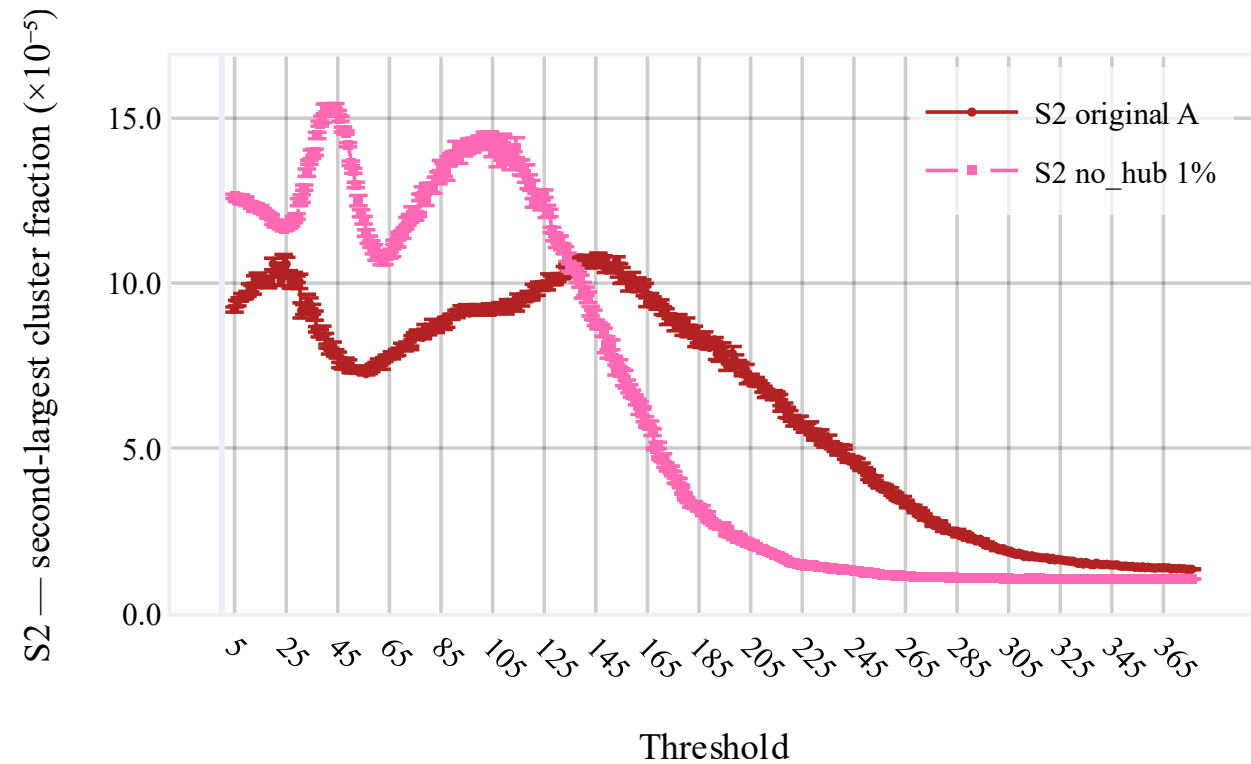


■ RESULTS: Greenberg-Hastings Model

➤ Drosophila Connectome

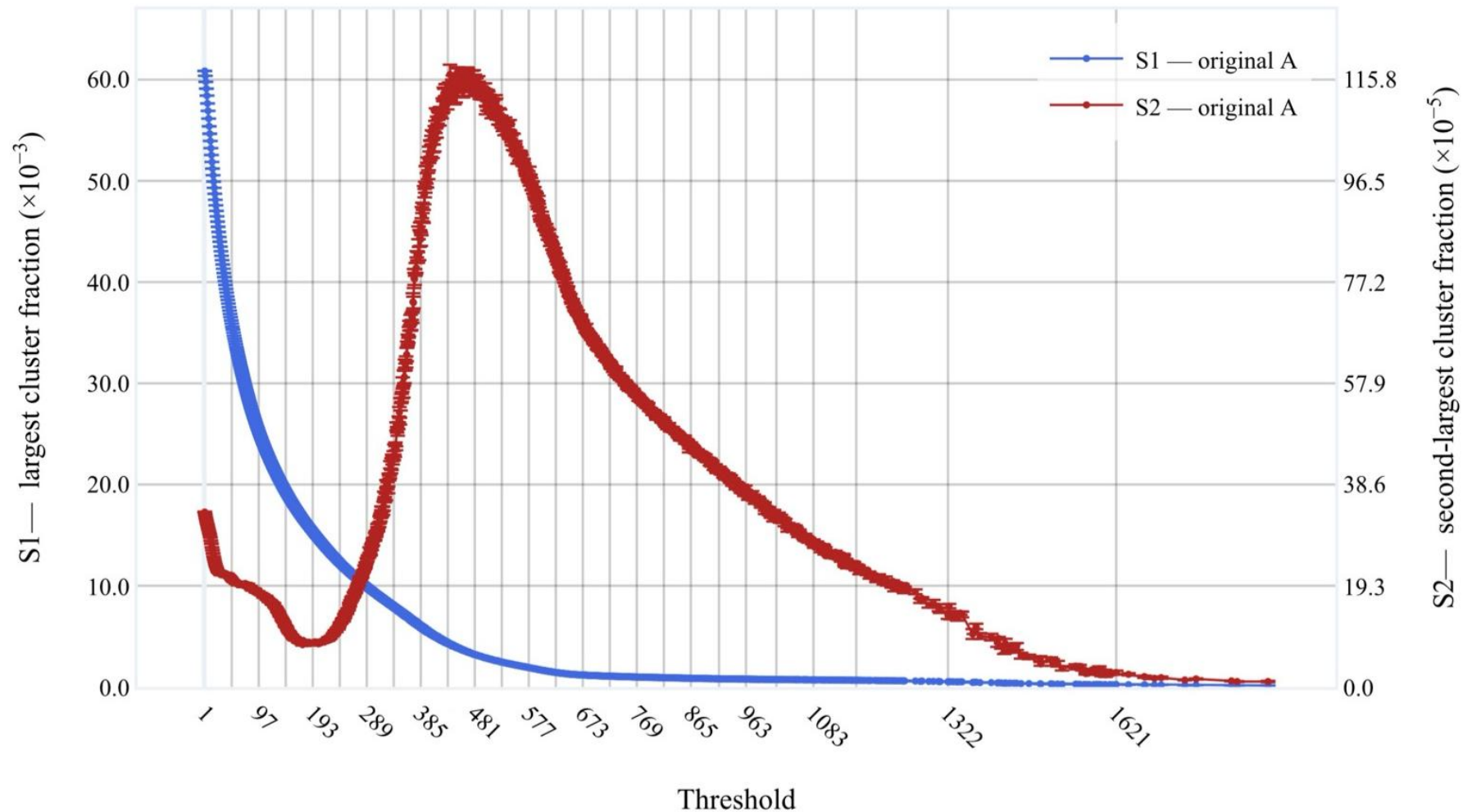
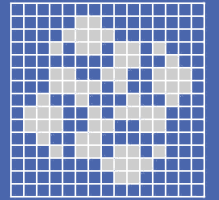


- S2 Comparison:
- Increase in Height
- Decrease in Width



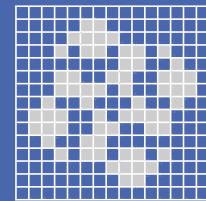
RESULTS: Greenberg-Hastings Model

➤ Human Connectome



■ RESULTS: Greenberg-Hastings Model

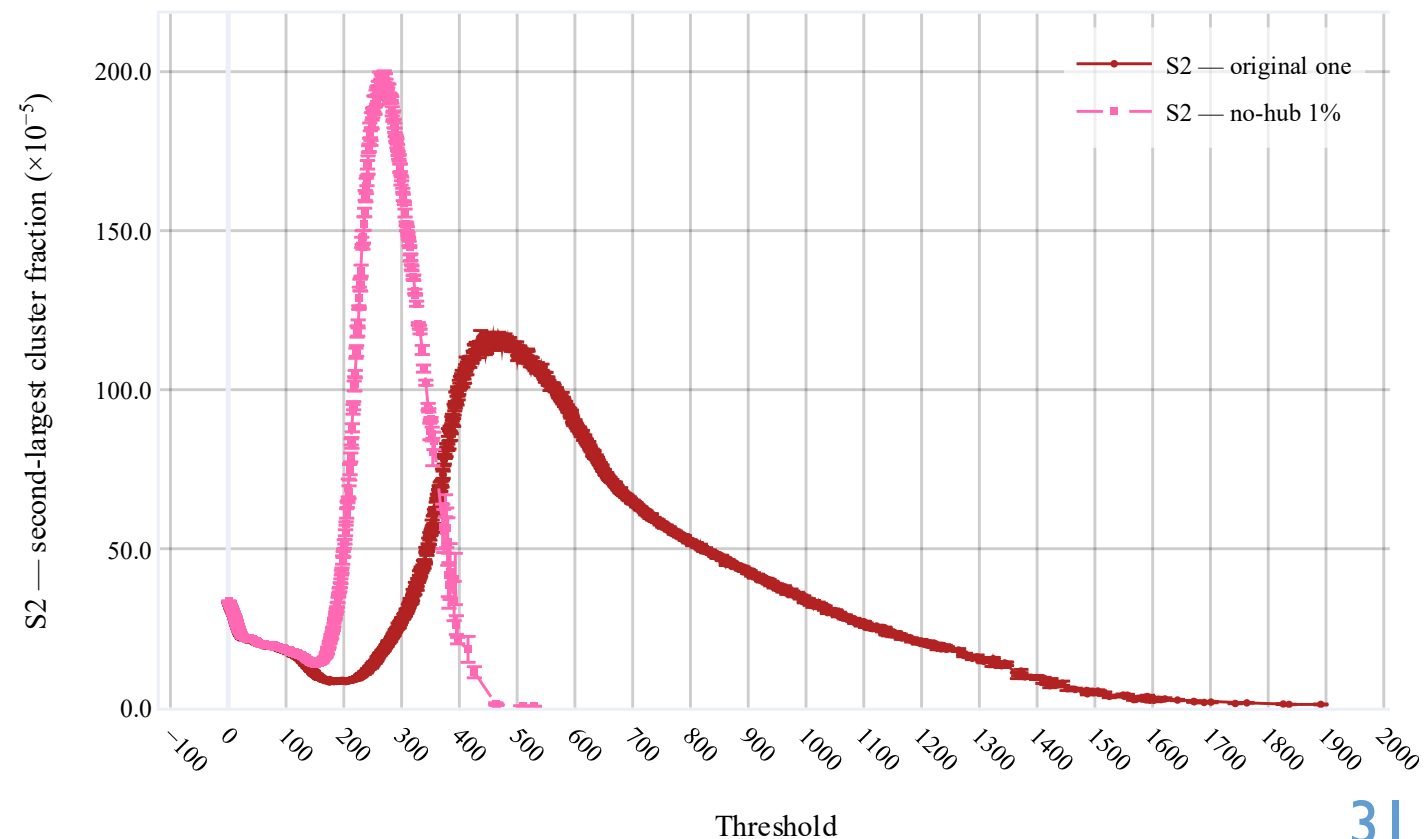
➤ Human Connectome



■ Comparison:

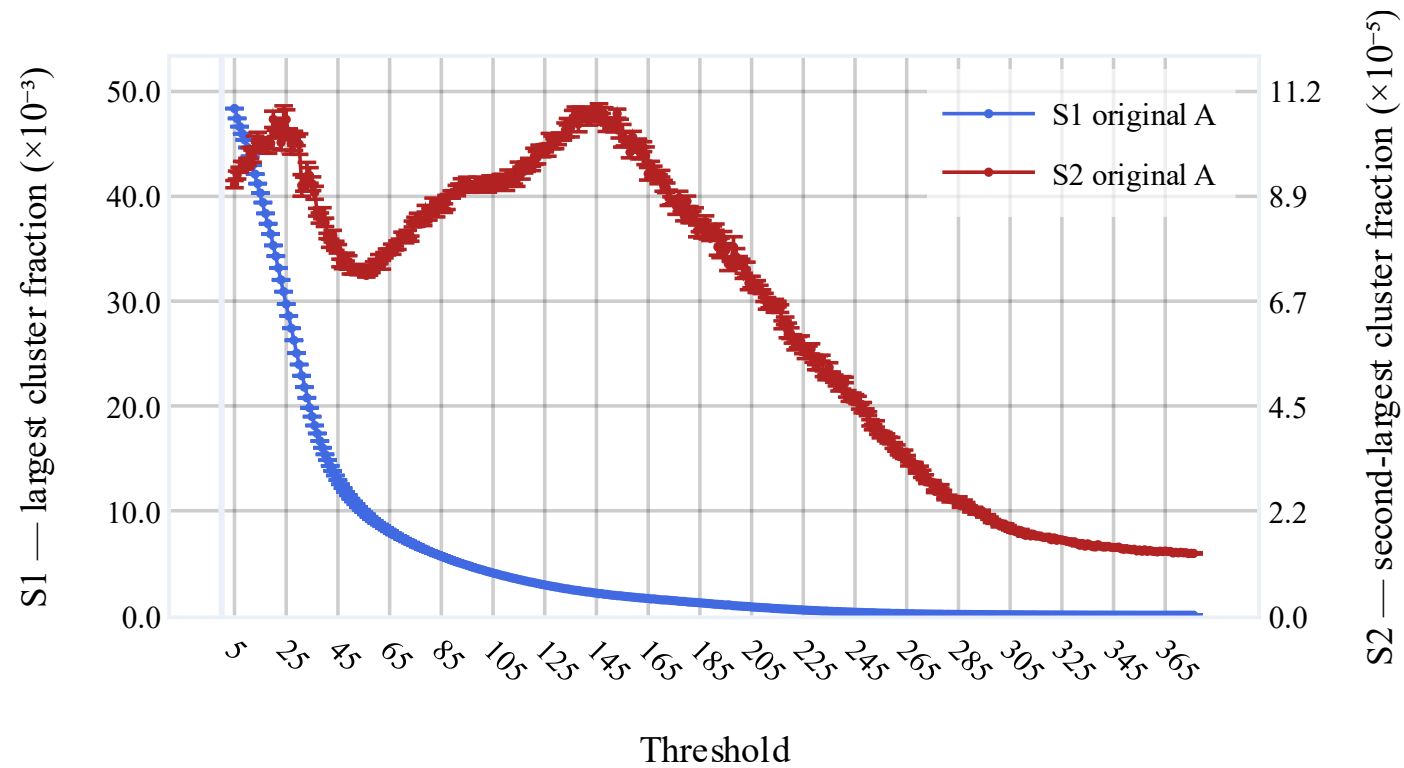
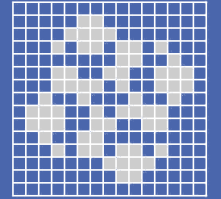
➤ Increase in Height

➤ Decrease in Width



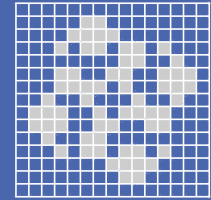
RESULTS: WHY 2 PEAKS?

➤ Drosophila Connectome



■ RESULTS: WHY 2 PEAKS?

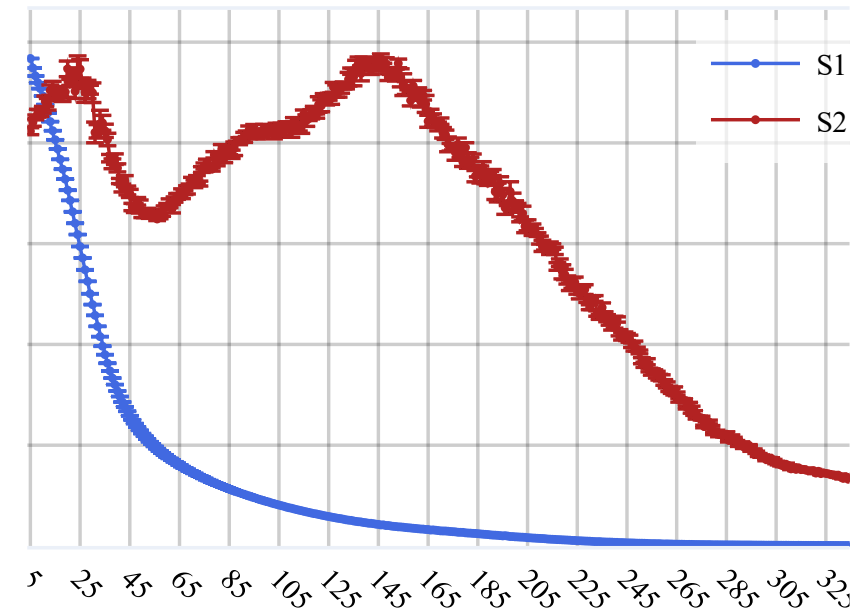
➤ Drosophila Connectome



Most Activated Degree in S1	In-degree	Out-degree	Total degree
First peak	13	17	30
Second peak	71	53	165

Most Activated Degree in S2	In-degree	Out-degree	Total degree
First peak	13	14	29
Second peak	40	42	79

Stats	In-degree	Out-degree	Total degree
Median	14	17	32
Third quartile	26	31	59



→ FUTURE WORK ←

- ✓ Structural Features Underlying 2 Peaks In The Drosophila S2-T Plot
- ✓ Inspecting the avalanche size distribution across the critical threshold range



SUMMARY

- Investigated critical behavior in neuronal populations
- Using connectomes
- How the system reaches the critical state?

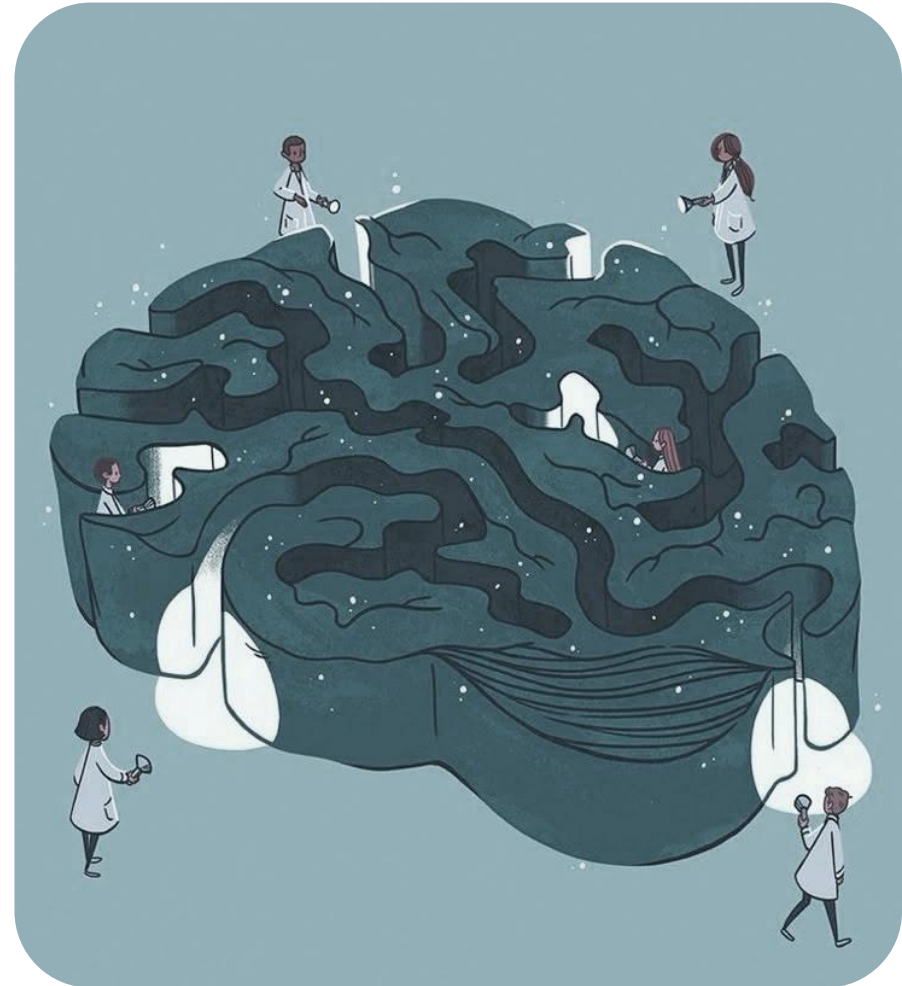
2 ideas:

→ self-organized criticality

🔍 tested by sandpile model

→ Griffiths phase

🔍 tested in by removing hubs in Greenberg-Hastings model



THANK YOU
FOR YOUR ATTENTION

